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Optical imaging lens

Abstract

An optical imaging lens includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a sixth lens in sequence along an optical axis from an object side to an image side. The first lens to the fifth lens are aspheric surface lenses and have diopters, and the sixth lens is a metalens and has a positive diopter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of China application serial no. 202210212731.7, filed on Mar. 4, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to an optical imaging lens.

Description of Related Art

(3) The specifications of portable electronic devices are ever-changing, and the optical imaging lens, one of the key components, is further developing diversely. How to make the lens of the portable electronic device have various focal lengths and provide high pixel and high resolution under the limited system length is a challenge to be faced and a problem to be solved.

SUMMARY

(4) The disclosure provides an optical imaging lens, which is provided with various focal lengths, good optical performance, and a small system length.

(5) According to an embodiment of the disclosure, an optical imaging lens is provided. The optical imaging lens includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a sixth lens in sequence along an optical axis from an object side to an image side. The first lens to the fifth lens are aspheric surface lenses and have diopters, and the sixth lens is a metalens and has a positive diopter.

(6) Based on the above, the optical imaging lens provided by the embodiment of the disclosure includes multiple lenses and a metalens. The metalens replaces the conventional multi-piece lenses with a binary pattern, greatly reducing the system length of the optical imaging lens. Moreover, the imaging lens is provided with various focal lengths and has good optical performance.

(7) To make the above-mentioned features and advantages of the disclosure more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a schematic diagram of an optical imaging lens according to an embodiment of the disclosure.

(2) FIGS. 1B to 1D are aberration curves of the optical imaging lens shown in FIG. 1A.

(3) FIG. 2 is a plan view of a metalens according to an embodiment of the disclosure.

(4) FIG. 3 is a plan view of a metalens according to an embodiment of the disclosure.

(5) FIG. 4 is a cross-sectional view of a metalens according to an embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

(6) FIG. 1A is a schematic diagram of an optical imaging lens according to an embodiment of the disclosure, and FIGS. 1B to 1D are various aberration curves of the optical imaging lens. Referring to FIG. 1A first, an optical imaging lens **10** of the embodiment includes a first lens **1**, a second lens **2**, a third lens **3**, an aperture **0**, a fourth lens **4**, a fifth lens **5**, a sixth lens **6**, and a filter **9** in sequence along an optical axis **I** of the optical imaging lens **10** from an object side **A1** to an image side **A2**. When the light emitted by an object to be photographed enters the optical imaging lens **10**, and transmits the first lens **1**, the second lens **2**, the third lens **3**, the aperture **0**, the fourth lens **4**, the fifth lens **5**, the sixth lens **6**, and the filter **9** in sequence, an image is formed on an image plane **99**. The first lens **1** to the fifth lens **5** are aspherical lenses and the diopters are negative, positive, negative, positive, and negative in sequence, and the sixth lens **6** is a metalens and has a positive diopter.

(7) The filter **9** is, for example, an infrared cut-off filter, which may allow light with an appropriate wavelength (e.g., infrared or visible light) to pass through, and filter out the infrared wavelength band to be filtered. The filter **9** is disposed between the sixth lens **6** and the image plane **99**. It shall be added that the object side **A1** is a side facing the object to be photographed, and the image side **A2** is a side facing the image plane **99**.

(8) In the embodiment, object side surfaces **15**, **25**, **35**, **45**, **55**, **65**, and **95** that face the object side **A1** and allow imaging light to pass through, and image side surfaces **16**, **26**, **36**, **46**, **56**, **66**, and **96**

that face the image side A2 and allow the imaging light to pass through are respectively provided on the first lens **1**, the second lens **2**, the third lens **3**, the fourth lens **4**, the fifth lens **5**, the sixth lens **6**, and the filter **9** of the optical imaging lens **10**.

(9) In the embodiment, the first lens **1** is a negative meniscus lens with a convex surface facing the object side A1, and both the object side surface **15** and the image side surface **16** of the first lens **1** are aspheric surfaces. The second lens **2** is a biconvex lens, and both the object side surface **25** and the image side surface **26** of the second lens **2** are aspheric surfaces. The third lens **3** is a biconcave lens, and both the object side surface **35** and the image side surface **36** of the third lens **3** are aspheric surfaces. The fourth lens **4** is a positive meniscus lens with a convex surface facing the image side A2, and both the object side surface **45** and the image side surface **46** of the fourth lens **4** are aspheric surfaces. The fifth lens **5** is a biconcave lens, and both the object side surface **55** and the image side surface **56** of the fifth lens **5** are aspheric surfaces.

(10) Regarding the metalens used as the sixth lens **6** (hereinafter referred to as the metalens **6**), please refer to FIGS. **1A**, **2**, **3**, and **4** at the same time, which show plan views (FIGS. **2** and **3**) or a cross-sectional view (FIG. **4**) of the metalens of embodiments of the disclosure, respectively. The metalens **6** includes a substrate **601** and multiple microstructures **602** disposed on the substrate **601**. The microstructures **602** are disposed on a first surface **601T** of the substrate **601**. A second surface **601D** of the substrate **601** is the image side surface **66** of the metalens **6**, and top surfaces **602T** of the microstructures **602** constitute the object side surface **65** of the metalens **6**. The metalens **6** further includes a center of symmetry C, and the microstructures **602** are disposed in a circularly symmetrical manner with respect to the center of symmetry C. The optical axis I passes through the center of symmetry C and is parallel to the normal of the substrate **601**. The metalens **6** replaces the conventional multi-piece lenses with the binary pattern formed by the above-mentioned microstructures **602** to provide the required phase retardation, and the system length of the optical imaging lens **10** is greatly reduced.

(11) In an embodiment of the disclosure, referring to FIGS. **2** and **4**, each of the microstructures **602** is a cylinder, and FIG. **4** may be regarded as a cross-sectional view along a line AA' of FIG. **2**. The microstructures **602** may have different widths in the direction perpendicular to the optical axis I as shown in FIGS. **2** and **4**.

(12) In another embodiment of the disclosure, referring to FIGS. **3** and **4**, multiple annular microstructures **602** and cylindrical microstructures **602** disposed at the center of symmetry C are disposed on the substrate **601**. The annular microstructures **602** are configured in a circularly symmetrical manner with respect to the center of symmetry C, and the center of symmetry C overlaps with the optical axis I. FIG. **4** may be viewed as a cross-sectional view along the line AA' of FIG. **3**. The microstructures **602** may have different widths in the direction perpendicular to the optical axis I as shown in FIGS. **3** and **4**.

(13) In the embodiment shown in FIGS. **2** to **4**, the width of the annular microstructure **602** or the cylindrical microstructure **602** in the direction perpendicular to the optical axis I may fall within the range of 50 nm to 230 nm. The height in a direction parallel to the optical axis I may fall within the range of 550 nm to 650 nm, which is 600 nm in an embodiment. The distance between the center of one cylindrical microstructure **602** and the center of another adjacent cylindrical microstructure **602** may fall within the range of 230 nm to 270 nm, which is 250 nm in an embodiment. In some embodiments, the distance between the center of one cylindrical microstructure **602** and the center of another adjacent cylindrical microstructure **602** may be less than or equal to 0.75λ , where λ is the wavelength of visible light. In some embodiments, the ratio of the height of the annular microstructure **602** or the cylindrical microstructure **602** in the direction parallel to the optical axis I to the width in the direction perpendicular to the optical axis I is greater than 2.5 and less than 12. In some embodiments, the annular microstructure **602** or the cylindrical microstructure **602** has a refractive index n, and n falls within the range of 2.4 to 2.7.

(14) Other detailed optical data of the embodiment are shown in Table 1. The overall effective focal

length (EFL) of the optical imaging lens **10** is 14.46 millimeter (mm), the half field of view (HFOV) is 9.76°, the system length (TTL) is 11 mm, the aperture value (F-number) is 3.40, and the image height is 2.5 mm. The TTL refers to the distance from the object side surface **15** of the first lens **1** to the image plane **99** on the optical axis I.

(15) TABLE-US-00001 TABLE 1 Optical Imaging Lens 10 EFL = 14.46 mm, HFOV = 9.76°, F-number = 3.40, Image height = 2.5 mm, TTL = 11 mm Radius of curvature Distance Refractive Abbe Element Surface (mm) (mm) index number Object infinity infinity First Object side 4.391 0.400 1.671 19.276 lens 1 surface 15 Image side 4.227 0.100 surface 16 Second Object side 3.293 1.804 1.545 55.987 lens 2 surface 25 Image side -8.110 0.100 surface 26 Third Object side -12.571 0.400 1.671 19.276 lens 3 surface 35 Image side 8.392 0.262 surface 36 Aperture 0 infinity 2.877 Fourth Object side -4.781 0.400 1.545 55.987 lens 4 surface 45 Image side -3.508 0.182 surface 46 Fifth Object side -4.405 0.400 1.671 19.276 lens 5 surface 55 Image side 72.410 2.211 surface 56 Sixth Object side infinity 0.700 1.458 67.796 lens 6 surface 65 Image side infinity 0.477 surface 66 Filter 9 Object side infinity 0.210 1.517 64.167 surface 95 Image side infinity 0.477 surface 96 Image infinity — plane 99

(16) The radius of curvature of the object side surface **15** shown in Table 1 refers to the radius of curvature of the object side surface **15** of the first lens **1** in the optical axis region, and the radius of curvature of the image side surface **16** refers to the image side surface **16** of the first lens **1** in the optical axis region, and so on. The distance of the object side surface **15** (0.400 mm as shown in Table 1) refers to the distance between the object side surface **15** and the next surface (the image side surface **16** in the example) on the optical axis I, that is, the thickness of the first lens **1** is on the optical axis I is 0.400 mm, and the distance of the image side surface **16** (0.100 mm as shown in Table 1) refers to the distance between the image side surface **16** and the object side surface **25** on the optical axis I, that is, the distance between the first lens **1** and the second lens **2** on the optical axis I is 0.100 mm, and so on.

(17) According to Table 1, the object side surface **15** of the first lens **1** is a convex surface and the image side surface **16** is a concave surface, the object side surface **25** of the second lens **2** is a convex surface and the image side surface **26** is convex surface, and the object side surface of the third lens **3** is a concave surface and the image side surface **36** is a concave surface, the object side surface **45** of the fourth lens **4** is a concave surface and the image side surface **46** is a convex surface, and the object side surface **55** of the fifth lens **5** is a concave surface and the image side surface **56** is a concave surface. The refractive indices of the first lens **1** to the fifth lens **5** fall within the range of 1.5 to 1.7, and the refractive index of the substrate **601** of the sixth lens **6** falls within the range of 1.4 to 1.5, which is 1.458 in the embodiment.

(18) In some embodiments of the disclosure, the optical imaging lens satisfies the conditional expression of $TTL/EFL < 0.8$, where TTL is the distance from the object side surface of the first lens **1** to the image plane **99** on the optical axis I, and EFL is the effective focal length of the optical imaging lens. In some embodiments of the disclosure, the optical imaging lens satisfies the conditional expression of $0.5 \times TTL < D1 < 0.75 \times TTL$, where D1 is the distance between the aperture **0** and the sixth lens **6** on the optical axis I.

(19) In the embodiment, the object side surfaces **15**, **25**, **35**, **45**, and **55** and the image side surfaces **16**, **26**, **36**, **46**, and **56** of the first lens **1** to the fifth lens **5** total ten surfaces and are all aspheric surfaces. These aspheric surfaces are defined according to the following formula (1):

(20) $Z(Y) = \frac{Y^2}{R} / (1 + \sqrt{1 - (1 + K) \frac{Y^2}{R^2}}) + \sum_{i=1}^n a_{2i} \times Y^{2i}$ (1) Y: the distance between a point on the aspheric curve and the optical axis I; Z: the depth of the aspheric surface, that is, the vertical distance between the point on the aspheric surface that is Y from the optical axis I, and the tangent plane tangent to the vertex of the aspheric surface on the optical axis; R: the radius of curvature of the lens surface; K: cone coefficient; a_{2i} : the 2i tier aspheric coefficient.

(21) The aspheric coefficients of the above-mentioned aspheric surfaces in formula (1) are shown

in Table 2 below. Among them, the column number 15 in Table 2 represents the aspheric coefficient of the object side surface **15** of the first lens **1**, the column number 16 represents the aspheric coefficient of the image side surface **16** of the first lens **1**, and the other columns follow the same analogy.

(22) TABLE-US-00002 TABLE 2 Surface K a.sub.4 a.sub.6 a.sub.8 a.sub.10 15 0 1.29E-03
-3.86E-05 1.44E-06 -6.17E-06 16 0 2.13E-03 1.17E-04 -2.39E-05 -8.72E-06 25 0
-1.18E-03 -7.56E-05 -3.36E-05 -5.45E-06 26 0 1.56E-03 -9.25E-06 -3.35E-05 4.00E-06
35 0 1.72E-03 2.31E-04 -2.29E-05 3.75E-06 36 0 -7.96E-04 2.29E-05 1.06E-05
-9.03E-06 45 0 -1.44E-02 -2.17E-03 -2.46E-04 9.30E-04 46 0 4.60E-03 -6.83E-03
-1.67E-03 7.45E-04 55 0 -3.08E-02 -7.17E-03 -1.45E-03 -1.20E-03 56 0 -3.48E-02
2.08E-03 -1.49E-03 1.47E-04

(23) The binary coefficients of the object side surface **65** and the image side surface **66** of the metalens **6** (the sixth lens **6**) are shown in Table 3.

(24) TABLE-US-00003 TABLE 3 Surface p.sup.2 p.sup.4 p.sup.6 p.sup.8 65 5.74E+04
-1.37E+04 1.51E+04 4.68E+05 66 -4.97E+04 -8.82E+03 -1.94E+04 4.95E+03

(25) Referring to FIGS. **1B** to **1D**, FIGS. **1B** and **1C** illustrate the field curvature aberration in the sagittal direction and the field curvature aberration in the tangential direction on the image plane **99** when the light wavelengths are 650 nm, 610 nm, 555 nm, 510 nm, and 470 nm, respectively. FIG. **1D** illustrates the distortion aberration on the image plane **99** when the light wavelengths are 650 nm, 610 nm, 555 nm, 510 nm, and 470 nm.

(26) In the two field curvature aberration diagrams in FIGS. **1B** and **1C**, the field curvature aberrations of the above five representative wavelengths in the entire field of view are within ± 0.06 mm, indicating that the optical system of the embodiment can effectively eliminate the aberrations. The distortion aberration diagram in FIG. **1D** shows that the distortion aberration of the embodiment is maintained within the range of $\pm 2\%$, which indicates that the distortion aberration of the embodiment meets the imaging quality requirements of the optical system. Therefore, compared with the existing optical lens, the embodiment can still provide good imaging quality under the condition that the aperture value is 3.40 and the image height is 2.5 mm, and the system length is only 11 mm.

(27) In summary, the optical imaging lens provided by the embodiments of the disclosure includes multiple lenses and a metalens. The metalens replaces the conventional multi-piece lenses with the binary pattern, greatly reducing the system length of the optical imaging lens. Moreover, the imaging lens is provided with various focal lengths and has good optical performance.

(28) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the claims appended hereto and their equivalents.

Claims

1. An optical imaging lens, comprising a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a sixth lens in sequence along an optical axis from an object side to an image side, wherein the first lens, the second lens, the third lens, the fourth lens, the fifth lens and the sixth lens respectively have a negative diopter, a positive diopter, a negative diopter, a positive diopter, a negative diopter and a positive diopter; the optical imaging lens has a total of six lenses; the first lens to the fifth lens are aspheric surface lenses, and the sixth lens is a metalens, wherein the optical imaging lens satisfies a conditional expression of $TTL/EFL < 0.8$, where TTL is a distance from an object side surface of the first lens to an image plane on the optical axis, and EFL is an effective focal length of the optical imaging lens.

2. The optical imaging lens according to claim 1, further comprising an aperture, disposed between

the third lens and the fourth lens.

3. The optical imaging lens according to claim 2, wherein the optical imaging lens satisfies a conditional expression of $0.5 \times \text{TTL} < D1 < 0.75 \times \text{TTL}$, where D1 is a distance between the aperture and the sixth lens **6** on the optical axis.
 4. The optical imaging lens according to claim 1, wherein the sixth lens comprises a substrate and a plurality of microstructures disposed on the substrate, and the plurality of microstructures are configured in a circularly symmetrical manner with respect to the optical axis.
 5. The optical imaging lens according to claim 4, wherein each of the microstructures is provided with a width W1 in a direction perpendicular to the optical axis and a width W2 in a direction parallel to the optical axis, and satisfies a conditional expression of $2.5 < W2/W1 < 12$.
 6. The optical imaging lens according to claim 5, wherein the width W1 falls within a range of 50 nm to 230 nm.
 7. The optical imaging lens according to claim 4, wherein the optical imaging lens satisfies a conditional expression of $D2 \leq 0.75\lambda$, where D2 is a distance between the microstructures adjacent to each other, and λ is a wavelength of visible light.
 8. The optical imaging lens according to claim 4, wherein the plurality of microstructures comprises a plurality of rings, and centers of symmetry of the plurality of rings overlap with the optical axis.
 9. The optical imaging lens according to claim 4, wherein the plurality of microstructures are disposed on the substrate in an array, and each of the microstructures is a cylinder.
 10. The optical imaging lens according to claim 4, wherein the plurality of microstructures have refractive indices n, and n fall within a range of 2.4 to 2.7.
 11. The optical imaging lens according to claim 1, wherein the first lens is a negative meniscus lens with a convex surface facing the object side.
 12. The optical imaging lens according to claim 1, wherein the fourth lens is a positive meniscus lens with a convex surface facing the image side.
 13. The optical imaging lens according to claim 1, wherein the second lens is a biconvex lens.
 14. The optical imaging lens according to claim 1, wherein the third lens and the fifth lens are biconcave lenses.
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